

Khizar Aamir

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EDUCATION

Bachelor of Science in Computer Science (Minor in Economics)

Sep 2022 – Dec 2026

University of British Columbia

Kelowna, BC

- **GPA: 4.2/4.33**; Dean's List (2023–2025), Deputy Vice-Chancellor Scholarship (2023, 2024)
- **Relevant Coursework:** Object Oriented Programming, Data Structures, Data Analytics, Web Programming, Machine Architecture, Software Engineering

EXPERIENCE

Software Engineer Intern

Dec 2024 – May 2025

Strike Den

Karachi, Pakistan (Remote)

- Engineered a **full-stack gym management platform** using **Next.js, Node.js, MongoDB, Tailwind CSS** with **JWT authentication** and **RBAC**, streamlining class scheduling and trainer management workflows.
- Built a **responsive admin dashboard** enabling staff to manage trainers, classes, and weekly scheduling with **real-time updates** and optimistic UI patterns.
- Implemented **secure API endpoints** with **CSRF protection** and **rate limiting**, optimizing MongoDB queries with **schema indexing** to achieve **257ms** median response times under **50** concurrent users with **99% reliability**.

Undergraduate Researcher — Directed Studies

Jan 2025 – May 2025

University of British Columbia

Kelowna, BC

- Developed an **OpenAI-driven simulation system** to emulate realistic student behavior for testing **sigmoid-based difficulty adaptation** and **mastery-weighted impact** algorithms in intelligent tutoring systems.
- Validated AI-generated data against human responses using **Kolmogorov–Smirnov tests** and **five-number summary analysis**, confirming **79%** distribution similarity with observed student behavior.
- Submitted a **short paper** to **CASCON 2025** detailing the reinforcement learning approach, statistical validation methods, and experimental findings.

Undergraduate Teaching Assistant & Subject Tutor

May 2024 – Dec 2024

University of British Columbia

Kelowna, BC

- Taught and supported **100+** students in **ECON 101** and **ECON 102** through one-on-one tutoring and structured problem-solving sessions.
- Assisted in lectures, tutorials, and grading, helping students master **introductory microeconomic and macroeconomic models** through targeted review and office hours.

PROJECTS

TSKFLO: Full-Stack Task Manager | *React, Redux, Node.js, Express, Socket.io, MongoDB*

Jan 2025 – April 2025

- Built a **collaborative task management platform** featuring **JWT authentication**, **real-time Socket.io messaging**, and a **Dockerized REST API** with **15+** endpoints for full CRUD operations.
- Implemented **role-based admin/user dashboards** with **automated task notifications**, reducing manual coordination by **70%** through streamlined workflows.

Loreleaf: Knowledge Management Platform | *Next.js, React, Node.js, PostgreSQL, Prisma*

Nov 2024 – May 2025

- Built a secure **Zettelkasten-inspired note-taking platform** with **JWT authentication**, **bcrypt hashing**, and **Zod validation**, achieving **sub-2ms API response times** via optimized queries and middleware.
- Developed an **interactive force-directed graph** using React Force Graph to visualize **bi-directional note links** with **real-time rendering**, enabling intuitive navigation of complex knowledge networks.

3D Interactive Portfolio: Developer Workspace | *React, React Three Fiber, Three.js*

Nov 2024 – Mar 2025

- Designed a 3D portfolio with **26 models**, modular React components, and smooth WebGL camera transitions.
- Added GLTF preloading and device pixel ratio capping for optimized performance on low-power GPUs.

ShakeScribe: AI-Powered Text Translator | *Python, Node.js, React, Flask, Redis, DeepSeek AI*

Dec 2024 – June 2025

- Developed an **AI-driven translator** using **DeepSeek LLM** and **all-MiniLM-L6-v2** model, implementing a **semantic similarity engine** that matches user input to **25 curated Shakespeare quotes** with cosine scoring and percentage accuracy.
- Architected a **full-stack application** with React/Tailwind frontend, Node.js/Express backend, and Python Flask analytics, integrating **Redis caching**, **100 req/day rate limiting**, and a **medieval-themed UI** with sonnet generation and real-time analytics.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL (MySQL, PostgreSQL), HTML/CSS, Swift

Frameworks & Libraries: React, Redux, Next.js, Node.js, FastAPI, SwiftUI, Tailwind CSS, Bootstrap, Pandas, NumPy

Tools & Platforms: Git, Docker, Firebase, Vercel, Postman, VS Code, Xcode, Android Studio, IntelliJ, Eclipse, MongoDB, Mongoose